

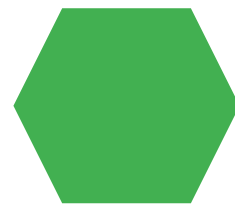


Seasonal Agriculture Performance Analysis

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PROBLEM STATEMENT

- Agricultural activities are influenced by seasonal variations in environmental conditions, farming practices, resource availability, and market conditions — causing performance to differ from one season to another.
- This project analyzes the given agricultural dataset to investigate seasonal differences in performance by identifying meaningful patterns, trends, relationships, and variations within the data.



Project Description

- This project analyzes a dataset of 4,000 farm records across 8 Indian states, covering three farming seasons — Kharif, Rabi, and Zaid.
- The analysis explores yield, profit, water usage, and environmental factors to understand how agricultural performance varies by season.
- It involves data cleaning, exploratory data analysis, visualization, and statistical testing (ANOVA) using Python.
- The project concludes with evidence-based recommendations to support better seasonal agricultural planning.

WHO ARE THE END USERS?

- Farmers planning seasonal crop selection
- Agricultural planners and government departments
- Agri-business companies and input suppliers
- Policy makers designing seasonal agriculture schemes
- Researchers studying seasonal farming trends



Technology Used

- Python
- Pandas & NumPy (data cleaning and analysis)
- Matplotlib & Seaborn (data visualization)
- SciPy (statistical testing – ANOVA)
- Jupyter Notebook / Google Colab



RESULTS

Data Loading & Initial Exploration



Dataset Preview (4000 rows × 28 columns)

Farm_ID	State	District	Crop	Season	Area(Ha)	Rainfall	Yield	Profit
SF10001	Andhra Pradesh	Rajkot	Wheat	Kharif	0.53	486.4	2.46	-11852
SF10002	Maharashtra	Nalgonda	Maize	Kharif	6.53	855.4	0.3	-451950
SF10003	Telangana	Warangal	Pulses	Rabi	4.86	455.6	0.52	-124744
SF10004	Telangana	Indore	Rice	Kharif	4.5	753.2	1.84	-95704
SF10005	Karnataka	Ludhiana	Maize	Zaid	4.21	101.6	2.21	-144352



RESULTS

Data Cleaning Summary (Missing Values & Duplicates)



Data Cleaning Summary

Rows: 4000 Columns: 28
Duplicate rows: 0
Missing values (before cleaning): Rainfall_mm=48, Soil_Moisture_pct=40, Yield_Tonnes_Ha=32
Missing values (after cleaning): 0



RESULTS

Season-wise Performance Summary



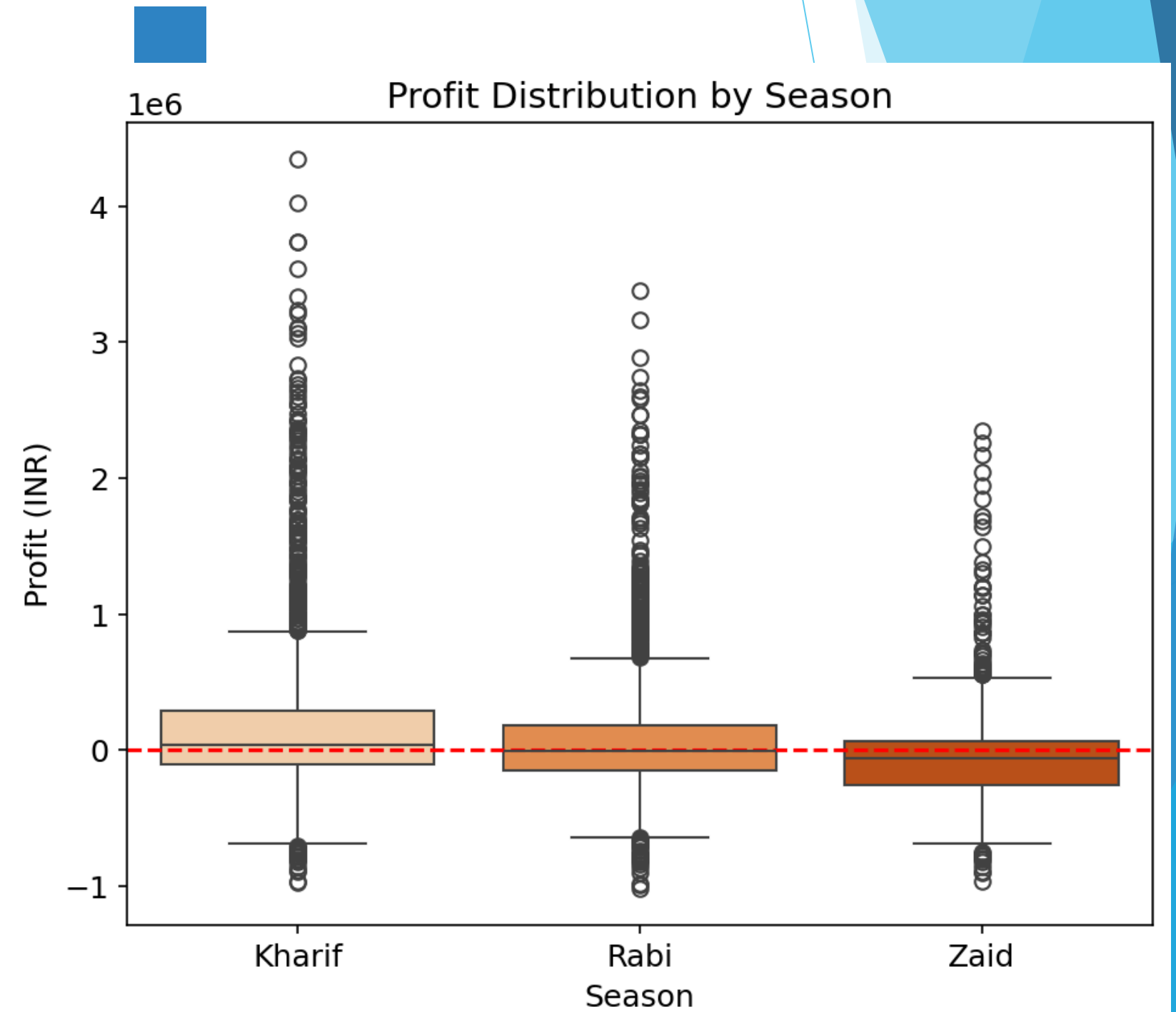
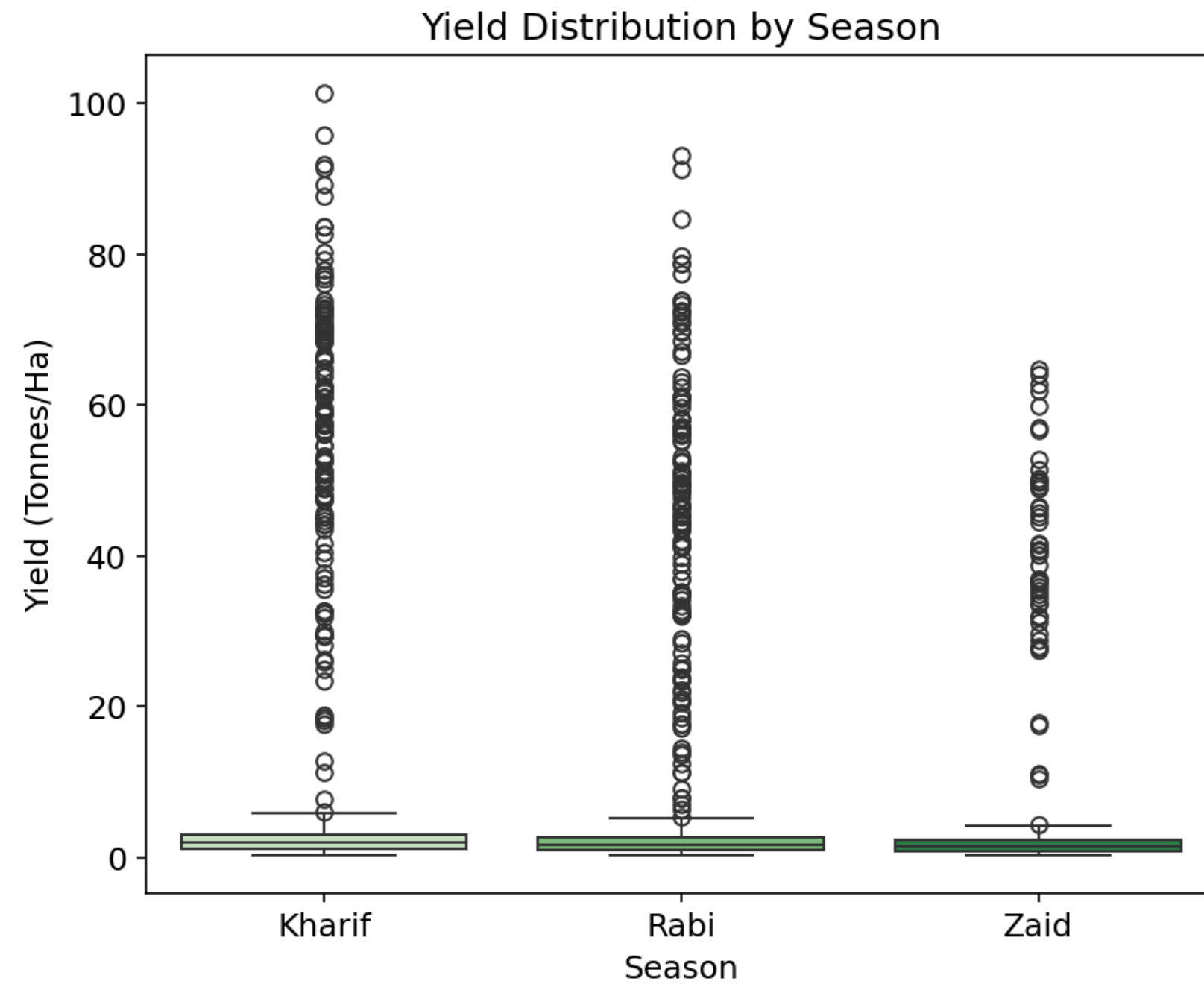
Season-wise Performance Summary

Season	Yield_Tonnes_Ha	Production_Tonne	Profit_INR	Revenue_INR	Total_Cost_INR	Water_Use_t_per_ha	Water_Efficiency_t_per_ha	Rainfall_mm	Disease_Pest_Risk
Kharif	5.63	46.31	178914.65	710719.06	531804.41	6102.2	5.89	852.11	54.47
Rabi	5.04	41.49	87689.47	601526.05	513836.58	5846.99	5.19	435.94	40.48
Zaid	4.64	38.89	-24804.82	519171.9	543976.73	6419.89	4.41	299.12	38.22



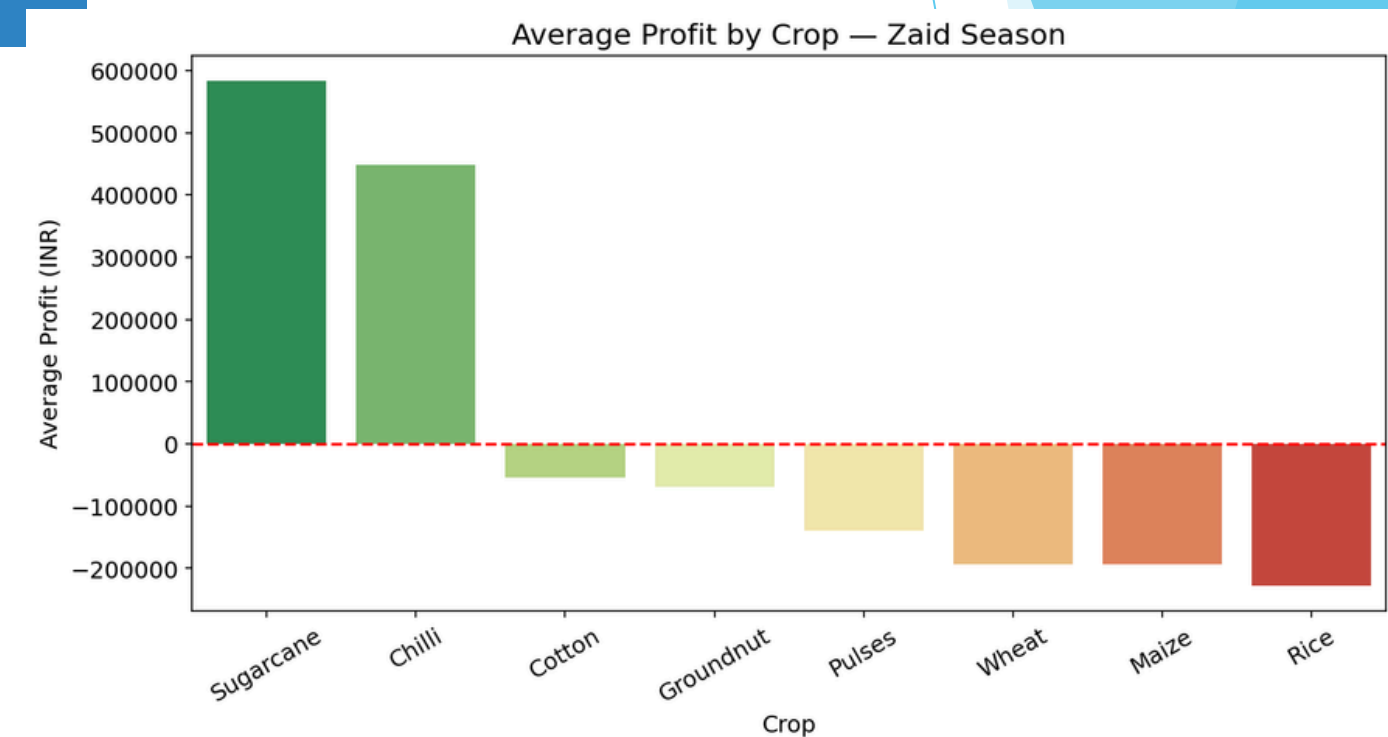
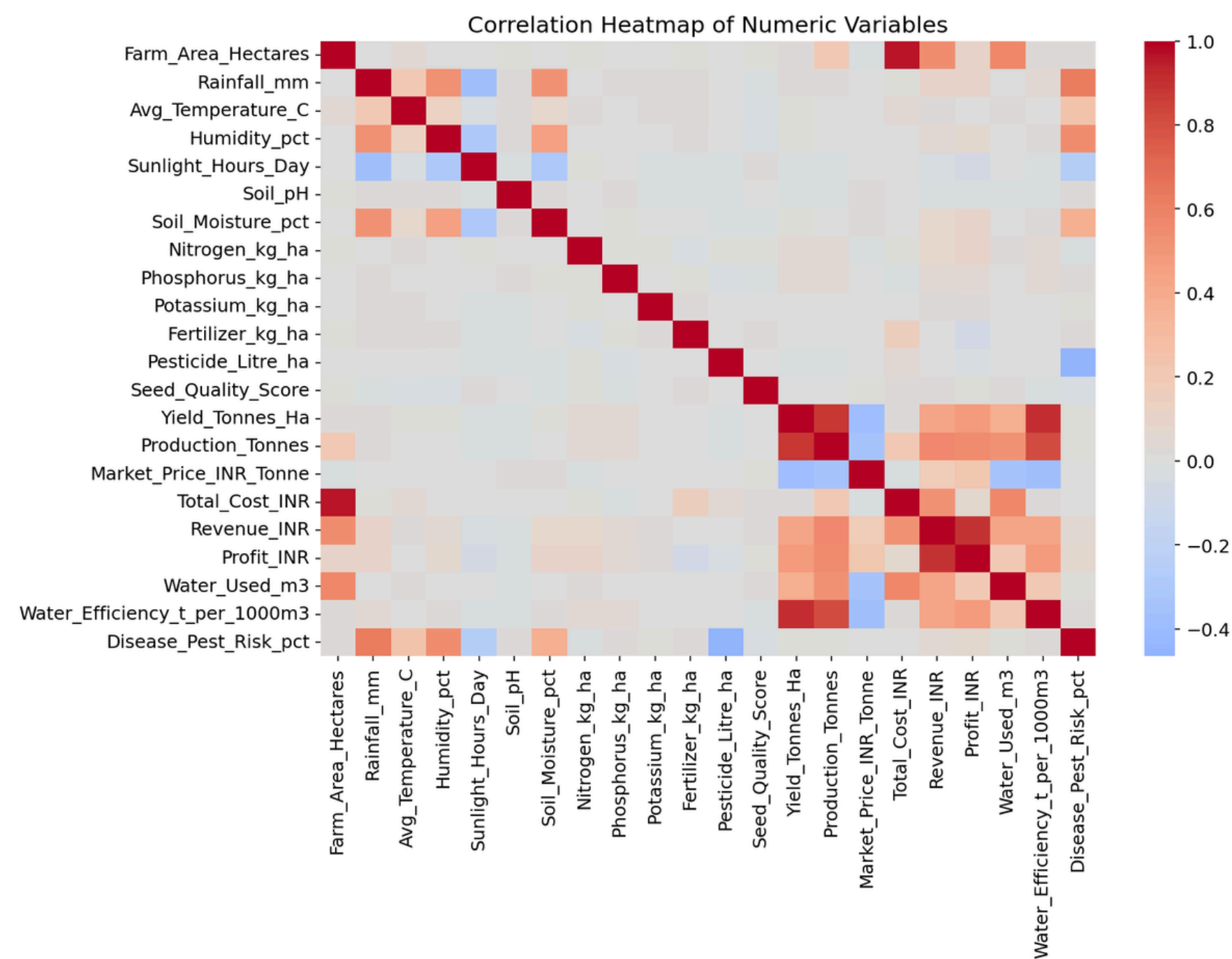
RESULTS

Yield & Profit Distribution by Season



RESULTS

Correlation Heatmap, Zaid Crop Profitability & ANOVA Test Results



ANOVA Statistical Test Results

Profit across Seasons: F-statistic = 34.29 P-value = 1.71e-15 -> Significant ($p < 0.05$)

Yield across Seasons: F-statistic = 1.55 P-value = 0.212 -> Not Significant ($p > 0.05$)

Future scope

- Predictive modeling to forecast profit and yield for upcoming seasons using machine learning.
- Crop recommendation system suggesting the most profitable crop for each season and region.
- Integration of real-time weather and soil sensor data for dynamic, up-to-date insights.
- Expansion of the dataset to cover more states, crops, and multiple years for stronger trend analysis.
- Market price forecasting to help farmers time their sales for maximum revenue.
 - An interactive dashboard for farmers and policymakers to explore seasonal insights easily.

GitHub Link

GitHub Repository:

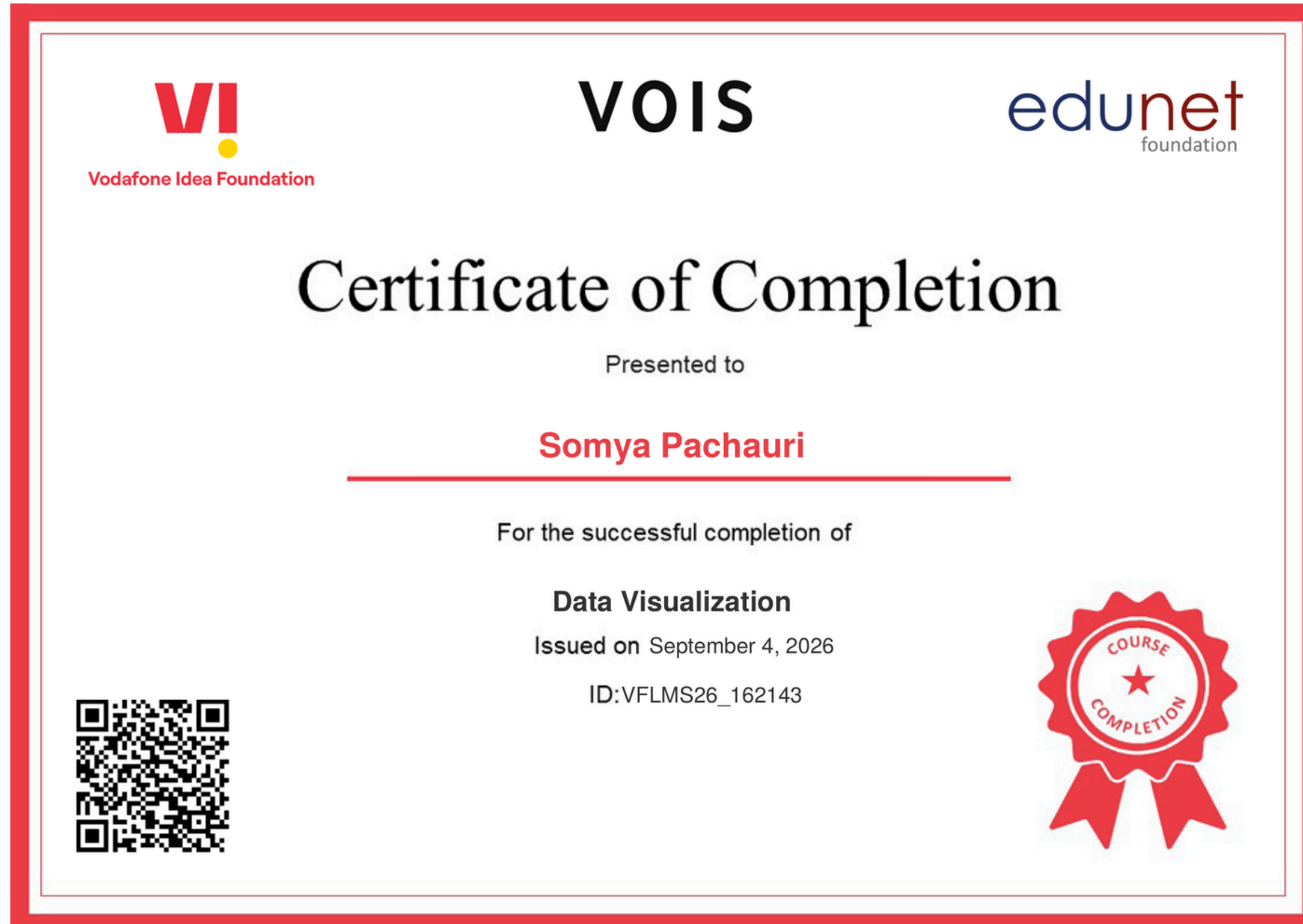


https://github.com/Somya1231code/Seasonal_Agriculture_Performance_Analysis

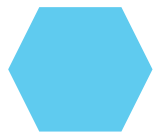


VOIS Course completion certificate -

Course name- Data Visualization



Thank You



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